



WHAT DOES LIMBIC MEAN?

The word "limbic" is derived from the Latin *limbus*, meaning "border," referring to the system's role as a kind of transition zone between the cortex and lower brain.

Reward and punishment

The limbic system is closely linked to feelings of rage and contentment. Both are due to the stimulation of reward or punishment centers within the limbic system, particularly in the hypothalamus. Reward and punishment are crucial aspects of learning, in that they create a basic response to experiences. Without this rating system, the brain would simply ignore old sensory stimuli that it had already experienced and pay attention only to new stimuli.



Pleasure

Associated with the release of dopamine, the brain seeks to repeat behaviors that create this feeling.



Disgust

This emotion is linked to the sense of smell. Its primordial role is to protect us from infection.



Fear

Fear is linked to specific stimuli by the amygdala. This can lead to a controlled rage or fight response.



HIPPOCAMPUS

Cingulate gyrus helps form memories associated with strong emotion



EPISODIC MEMORIES

The hippocampus receives and processes inputs from the cerebrum. It is involved in creating episodic memories, or memories about what you have done, and creating spatial awareness.

Klüver-Bucy syndrome

This condition is caused by damage to the limbic system and results in a spectrum of symptoms associated with the loss of fear and impulse control. First described in humans in 1975, this neural disorder is named after the 1930s investigators Heinrich Klüver and Paul Bucy, who performed experiments that involved removing various brain regions in live monkeys and noting the effects.

In humans, the syndrome may be caused by Alzheimer's disease, complications from herpes, or brain damage. It was first documented in people who had undergone surgical removal of parts of the brain's temporal lobe. The condition can be treated with medication and assistance with daily tasks.

SYMPTOM	DESCRIPTION
Amnesia	Damage to the hippocampus leads to the inability to form long-term memories.
Docility	With little sensation of reward for actions, sufferers lack motivation.
Hyperorality	An urge to examine objects by putting them in the mouth.
Pica	Eating compulsively, including inedible substances like earth.
Hypersexuality	A high sex drive often associated with fetishes or atypical attractions.
Agnosia	Losing the ability to recognize familiar objects or people.